Computer Networks
Fall 2024/25

Exercise-5

Submission in pairs by Thursday, **30-1-2025.** Submission is done by uploading your work to the course Moodle website. >>> No late submissions will be accepted
The name of the submitted file must be Exercise5_username1_username2**.pdf**
For example: Exercise4_israel.israeli_david.davidi.pdf (with dots). The first and last names of both students must appear. Only one student should submit the assignment (not both). It Covers Chapter 5 from the textbook. Please notice that once the grades for an exercise are published, appeals can only be submitted in the following five days. The grader is Mr. Daniel Ohayon, and you may contact him by email at *reichman.computer.networks@gmail.com.*

Problem 1 (25 points – Lecture 10).
Consider the scenario where three hosts, with IP addresses 10.1.1.1, 10.1.1.2, and 10.1.1.3, are in a local network behind a NATted router (i.e., NAT is implemented on the router) that sits between these hosts and the larger Internet. IP datagrams being sent from, or destined to, these three hosts from the Internet must pass through this NAT router. The router's interface on the LAN side has the IP address 10.1.1.55, while the router's address on the Internet side has the IP address 41.42.43.44. Suppose the host with IP address 10.1.1.1 sends a packet destined to server 200.82.83.84 (on the Internet side). The source port is 3333, and the destination port is 8080. After the server gets the packet, it sends a response (one packet). In each of the four steps shown below (which follow each other in time), what are the packet's source and destination IP addresses and the packet's source and destination port numbers?

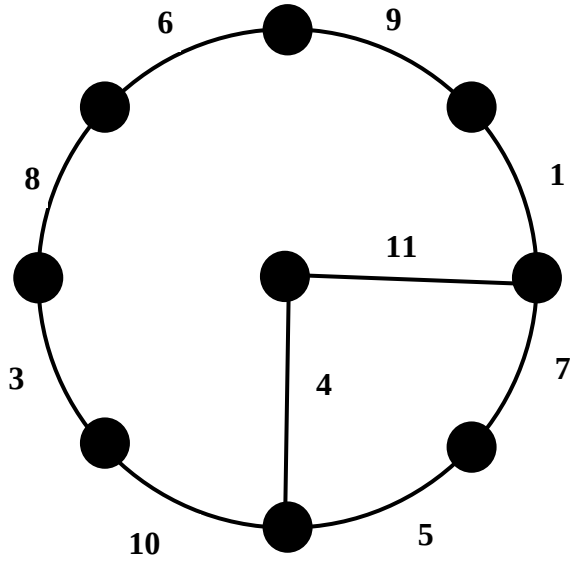
Step	Source IP	Destination IP	Source Port	Destination Port
1 – from host				
2 – from NATted router			**6688**	
3 – from server				
4 – from NATted router				

Specify below the entry that has been made in the NAT table.

Internet side	LAN side

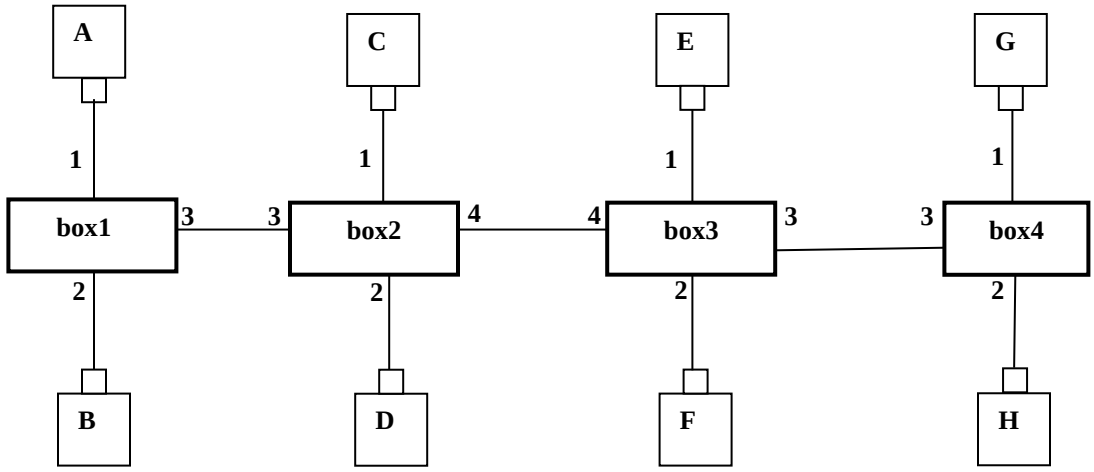
Problem 2 (25 points – Lecture 11).

The question refers to the distributed algorithm for minimum-weight spanning tree (MWST).
Assume that you execute the algorithm on the graph in the picture below until it terminates.
1. How many edges will become a **core** during the execution, and what are these edges**?**
2. What will be the **level** of the (single) fragment when the algorithm terminates?
Briefly justify the answers.
Remark: The answer is the same for all possible executions.



Problem 3 (25 points).
Consider the network below. Suppose all the routing tables are up to date. The four boxes, box1, box2, box3, box4 are either switches (i.e., bridge) or hubs as specified later.
Assume that A sends a packet to B and B responds to A. After that, C sends a packet to D, and D responds to C. After that, E sends a packet to F, and F responds to E. After that, G sends a packet to H, and H responds to G.
Assume that in an ARP query, **both** the sender **and** the receiver update their ARP tables when necessary.

1. Assume that box1 and box4 are switches and box2 and box3 are hubs and the switch tables are empty. Show the tables of each one of the boxes after the packets are received, assuming that the ARP tables are empty. Write the entries in the tables **in the order** in which they are updated.
2. Assume that box1 and box4 are switches, box2 and box3 are hubs, the switch tables are empty, and the ARP tables are empty. How many different packets (both queries, responses, and IP packets) are sent?
3. Assume that box1 and box4 are switches, box2 and box3 are hubs, and both the switch tables and the ARP tables are up to date. How many packets (of all types) pass through the link between box2 and box3 (in both directions together)?
4. Assume that box2 and box3 are switches, box1 and box4 are hubs, the switch tables are empty, and the ARP tables are up to date. How many packets (of all types) pass through the link between box2 and box3 (in both directions together)?
5. Assume that box1 and box4 are switches, box2 and box3 are hubs, the switch tables are empty, and the ARP tables are up to date. How many packets (of all types) pass through the link between box2 and box3 (in both directions together)?
6. Assume that box2 and box3 are switches, box1 and box4 are hubs, the switch tables are up to date and the ARP tables are empty. How many packets (of all types) pass through the link between box2 and box3 (in both directions together)?
7. Assume that box2 and box3 are switches, box1 and box4 are hubs, and both the switch tables and the ARP tables are up to date. How many packets (of all types) pass through the link between box2 and box3 (in both directions together)?
Briefly justify your answers to 2-5 questions.



Problem 4 (25 points).
Solve Problem 3 under the assumption that in an ARP query, **only** the sender updates its ARP tables when necessary (the receiver does not update!).